

Backend Schema

Database: PostgreSQL (with PostGIS enabled)

1. farmer_profiles

- `id` (UUID, PK)
- `phone_number` (VARCHAR, Unique)
- `full_name` (VARCHAR)
- `total_land_ha` (NUMERIC)
- `cattle_count` (INT)

2. buyer_profiles

- `id` (UUID, PK)
- `phone_number` (VARCHAR, Unique)
- `shop_name` (VARCHAR)
- `shop_type` (VARCHAR - Enum)
- `sourcing_radius_km` (NUMERIC)
- `shop_location` (GEOGRAPHY(POINT, 4326))

3. crop_listings

- `id` (UUID, PK)
- `farmer_id` (UUID, FK -> farmer_profiles)
- `crop_name` (VARCHAR)
- `quantity_kg` (NUMERIC)
- `calculated_price_per_kg` (NUMERIC)
- `status` (VARCHAR - Enum: AVAILABLE, LOCKED, SOLD)
- `location` (GEOGRAPHY(POINT, 4326))
- **Index:** GIST index on `location` for hyper-fast spatial lookups.

4. farm_ledger

- `id` (UUID, PK)
- `farmer_id` (UUID, FK -> farmer_profiles)
- `transaction_type` (VARCHAR - Enum: INCOME, EXPENSE)
- `amount` (NUMERIC)
- `category` (VARCHAR - e.g., 'Seeds', 'Water', 'Crop Sale')
- `timestamp` (TIMESTAMPTZ)